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Statistical Models and Data Science Concepts

Unit D - Probability and Statistics

2024

Overview

Unit A. The basics of probability and random variables.

Unit B. **Parametric probabilistic models of** random variables

Unit C. **Estimation of unknown parameters** (point and CI)

(in random samples and from the basics of statistical inference)

Unit D. Information from **parametric statistical models**

- Using R to process data
- Emphasising the interpretation and predictive capacity of the model
- Introducing models linked to data science (e.g., machine learning)

Contents

Statistical studies

Statistical tools

Confidence intervals, hypothesis tests and Neyman Pearson lemma

STATISTICAL MODELS. Notation, cases, transformations and R functions

(a) A model on a specific value of a parameter
(μ of a sample or μ_D of the difference in paired samples)

(b) A model comparing the parameter μ in independent samples

(c) A simple multiple LINEAR model
(coefficients of a line as parameters β_0 β_1 ...)

\+ Other techniques (data science: multidimensional visualisation, clustering, machine learning, data mining, etc.)

Statistical studies

Statistical studies obtain **information** from the **data** by analysing the relationship between variables. The study must describe the data (**descriptive statistics**).

As we have seen in block C, we must decide:

- how to obtain the data
  Only **random** samples justify uncertainty measures such as standard error (SE) and confidence intervals.
- the role of each variable
  Response ***Y***, assignable cause ***X***, and covariates ***Z***.
- the types of study
  Experimental or observational.
- the design
  Paired, independent groups, etc.

To be able to increase knowledge, a **scientific** study must be **reproducible**.

Research fails if it cannot be reproduced.

The inability to reproduce an experiment is a scientific and social problem:

Methodologically poor research is a waste of resources.

Now we'll see some **tools** (some already presented in unit C) and statistical **models** for representing relationships between variables.

Statistical tools. CIs, tests and NP lemma

- Statistical inference is used to infer characteristics of the population (parameters) from the observations of a sample in order to estimate parameters or make decisions.
- A **confidence interval (CI)** is used to estimate the magnitude of an effect (mainly in an experimental study) and informs of the likely values of a parameter
- We may also want to compare a specific value of the parameter (a hypothesis) with the data. Significance tests (**tests,** hypothesis tests, HTs) evaluate the **evidence** against a claim This is a conservative methodology that proposes a null hypothesis (H_0) as a starting point (claim) and studies whether the data provide evidence against it. *Failing to reject H_0 does not mean that H_0 is true* (absence of evidence does not equal evidence of absence)
- We can focus on **making decisions** by limiting risks and comparing two hypotheses. The **Neyman-Pearson lemma** identifies and minimises errors in deciding on one of the two options.

In Appendix D of this unit you will find more information about HTs and the types of error in NP decisions.

The Reading List (Statistics for Computer Engineers) offers more details of significance and contrast tests of in Chapter 4.

Statistical tools and P-value

- A CI shows, for a certain risk, **certain values for a parameter** according to the empirical evidence from data, using a **statistic** with a known distribution
- **P-value** is the probability to obtain a value more extreme than the one obtained with the **statistic evaluated using a specific value** of the parameter in the observed sample

A "small" P-value shows small probability that the specific value would be credible according to the empirical evidence from data:

- (small probability to find another sample more "extreme")
- (difficult to justify that the difference between the sample and the specific value is due to hazard).

If the P-value is not "small" it doesn't evidence that the real value is the specific one evaluated

Common Fallacy: a large P-value means that the hypothetical parameter value should be correct

The Reading List (Statistics for Computer Engineers) offers more details in Chapter 4.

Results in R are "Estimate", "Std.Error", and the **statistic** (for example "t" o "t value") with the **P-value** (for example "Pr(>|t|)" o "p-value")

STATISTICAL MODELS. Notation

A statistical model combines a deterministic part (a formula with **parameter/s θ**) with a random part (with an appropriate distribution $\boldsymbol{\varepsilon}$) to explain the outcome *Y*:

$$\boldsymbol{Y_i = f(\theta) + \varepsilon_i}$$

$\boldsymbol{f(\theta)}$ is the deterministic part that will explain the value of the response *Y* based on parameters (μ for the central tendency, coefficients of a straight line, etc.) from which we obtain point estimates and confidence intervals. For example:

- the value of the central tendency parameter (mean) in a sample of the response variable *Y*
- the difference between means of two (paired or independent) samples of the response variable Y
- the equation of a straight line that relates two variables in a sample, *Y* and *X*

$\boldsymbol{\varepsilon_i}$ is the part corresponding to chance, and it should be preferably residual

- It does not inform about the relationship between the variables, it's different in each observation
- a small variability is preferable,
- and typically it's a Normal Distribution with mean equal to 0 ($\varepsilon \sim N(0, \sigma)$)

Using the model and the characteristics of an observation *i* we can obtain:
- **predictions**, $\widehat{Y}_i = f(\hat{\theta})$ applying the estimators of the deterministic part of the model
- **errors** or **residuals** ($\mathbf{e_i}$) subtracting predicted values ($\widehat{Y}_i$) from the observed ones (Y_i)

STATISTICAL MODELS. Cases

a) A model on a specific value of the **parameter μ**

(prediction *i*)

$\mathbf{Y_i = \mu + \varepsilon_i}$

The aim is point estimate (and CI) of μ ($\hat{\mu} = \bar{y}$) $(\hat{Y}_i = \hat{\mu} = \bar{y})$

Particular case: the **parameter μ in paired samples** (Y_1,Y_2)

$\mathbf{D_i = \mu_D + \varepsilon_i}$ D = Y1 – Y2 (difference as a response variable)

The aim is point estimate (and CI) of μ_D ($\hat{\mu}_D = \bar{d}$) $(\hat{D}_i = \hat{\mu}_D = \bar{d})$

b) A model comparing the **parameter μ** in **k** independent **groups** ($k \geq 2$)

$\mathbf{Y_i = \mu + \vartheta_j + \varepsilon_i}$

The aim is point estimate (and CI) of each μ_j ($\hat{\mu}_j = \bar{y} + \hat{\vartheta}_j$) of group *j* $(\hat{Y}_{ij} = \hat{\mu}_j = \bar{y} + \hat{\vartheta}_j)$

(μ as the reference mean and ϑ_j as the change to group *j*)

c) A **simple** or **multiple linear** model

$\mathbf{Y_i = \beta_0 + \beta_1 X1_i + \varepsilon_i}$ $(\hat{Y}_i = \hat{\beta}_0 + \cdots + \hat{\beta}_k Xk_i)$

$\mathbf{Y_i = \beta_0 + \beta_1 X1_i + \beta_2 X2_i + \cdots + \beta_k Xk_i + \varepsilon_i}$ $= b_0 + b_k Xk_i)$

The aim is point estimate (and CI) of coefficients β_i ($\hat{\beta}_0 = b_0$, $\hat{\beta}_1 = b_{1,\ldots}$) of the equation

The relationship between the response and the explanatory variables is a linear relation

+ Other techniques (data science: multidimensional visualization, clustering, machine learning,...)

For example: PCA [Y1, Y2, ... Yk] ---- geometrical *transformation* ----> $[\Psi 1, \Psi 2 \ldots] [\ldots \Psi k]$

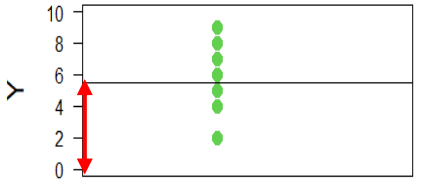
STATISTICAL MODELS. Function lm() in R

(lm from "*linear model*" but usually case c) is the one we call linear model)

Function lm() in R for the previous different cases:

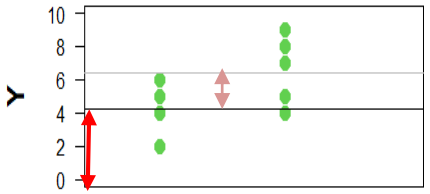
a) $Y_i = \mu + \varepsilon_i$ `lm(Y~1)`

$D_i = \mu_D + \varepsilon_i$ `lm(D~1)`



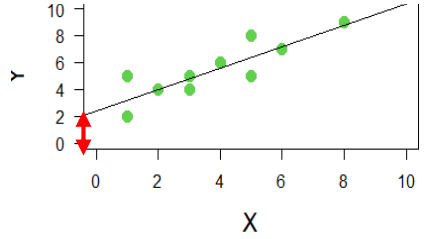
b) $Y_i = \mu + \vartheta_k + \varepsilon_i$ `lm(Y~G)`

`(G column separating k grups)`



c) $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ `lm(Y~X)`

`(` or `lm(Y~X1+X2+...+Xk))`



R results (**parameter point estimations**) :

- **"Coefficients"** are the **"Estimates"** for the corresponding parameter:
  - $\hat{\mu}$ or $\hat{\mu}_D$ or $\hat{\beta}_0$ as the reference values of the response Y (↕)
  - $\hat{\vartheta}_j$ as the change in means between groups (↕)
  - $\hat{\beta}_1$ as the slope of the equation corresponding a linear relation

Some more results for each point estimation:

- **"Std Error"** is the standard error that we can use to calculate CI
- **"p-value"** to evaluate the credibility to specific values of the parameter (0 is the default value)

STATISTICAL MODELS. Evaluation

To evaluate models we will perform:

- **Validation of assumptions** to confirm that the model is acceptable

  (**random variable** and **normality** will be fundamental assumptions apart from some specific ones)

- **Analysis of results** in several aspects:
  - Indicators to evaluate the "quality" of results and the **predictive capacity of models**
  - **Interpretation** of results

Model evaluation is necessary to show if the model is suitable and fit for purpose, or if the model could have **overfitting** (some more information at the end of this unit D)

In next sections we will see in detail the previous models with specific characteristics, specially for **validation of assumptions** and **analysis of results**

But, before these model sections, we will see how to deal with some necessary transformations to data (e.g. logarithms) and then how to concern to initial data

STATISTICAL MODELS. Transformations

Sometimes a **transformation** (often **logarithmic**) is needed to fulfill the assumptions (e.g., normality).

Then you have to **undo* the transformations in the estimates** obtained (point or using CI).

- In case of models for parameter μ, if normality is satisfied by ln(Y), the we use

  ln(Y) = Y' = μ' + ε obtaining the estimations $\bar{y}'$ and IC = [inf',sup']

  and undoing logarithms **exp($\bar{y}'$)** and **[exp(inf') , exp(sup')]** estimations for the initial response

  - In case of paired samples, the difference could have negative values. If the difference increases proportional to Y, let's try the difference in logarithms or the logarithm of the ratio Y1/Y2.

    ln(Y1) - ln(Y2) = ln(Y1/Y2) = Y'' = μ'' + ε with estimations $\bar{y}''$ i IC = [inf'',sup'']

    and undoing logarithms **exp($\bar{y}''$)** and **[exp(inf'') , exp(sup'')]** estimations for the initial response

- In case of linear model, the fitted equation is calculated with the transformed variables and the we can undo the transformation to predicted values

* Undoing the logarithmic transformation we obtain an CI for μ' not for μ (μ is **arithmetical** mean, and μ' is **geometrical mean**)

Statistical models
Model (a) and functions in R
(model about the value of parameter μ)

Model to estimate μ

Notation for the model: $Y_i = \mu + \varepsilon_i$

Functions in R: `lm(Y~1)` and `summary(lm(Y~1))`

(constant `1` means in R that we want to estimate only the parameter **μ**)

R outputs:

- point estimation of μ ($\hat{\mu} = \bar{y}$): **"Estimate"** of **"intercept"**
- standard error estimation: **"Std. Error"** (*se*) of **"intercept"** ($\bar{y}$). It allows to calculate CI

$$IC(1-\alpha\%, \mu) = \bar{y} \pm t_{n-1,1-\alpha/2} \cdot se$$

- value for the statistic (**"t value"**) and **p-value** (**"Pr(>|t|)"**) to evaluate the credibility for a possible parameter value (0 by default)
- standard deviation of residuals (**"Residual Standard error"**) unexplained by the model

Model to estimate μ

Validation of assumptions

Assumptions are: **random variable** and **normality of the response**

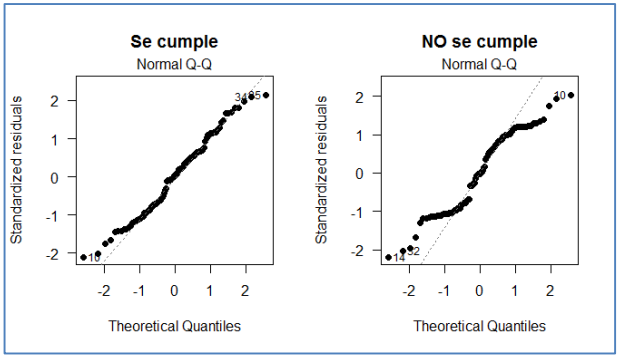
- **Random variable** assumption is not possible to verify. It depends on the sample design.
  It is only possible to assure independency in the order of the observations

- **Normality** assumption will be verified graphically with a **Normal QQ-plot**

  "Normal QQ plot" plots theoretical quantiles in front of sample quantiles

  When the plot is aligned there is evidence for assuming normality

  R functions are **`qqnorm(Y)` `qqline(Y)`**



Analysis of results

The main result is point estimation (and CI) for the population mean

The Residual standard error quantifies the deviation unexplained by them model

Model to estimate μ. Example

Sample of 9 positive and negative values (for example measuring errors)

```
> X = c(-4,-2,-1, 0, 0, 4, 8, 8, 9) # mean=2.4 sd=4.9
```

Unit C solution (CI)

```
> t.test(X)
```

```
t = 1.496, df = 8, p-value = 0.173
  alternative hypothesis: true mean is not equal to 0
  95 percent confidence interval:
   -1.323423  6.212312
  sample estimates: mean of x  2.444444
```

Unit D solution (statistical model)

```
> (mod <- lm(Y~1))
> summary(mod)
```

```
Residuals:
   Min     1Q Median     3Q    Max
-6.444 -3.444 -2.444  5.556  6.556

Coefficients: Estimate  Std.Error   t value  Pr(>|t|)
(Intercept)     2.444      1.634     1.496    0.173
Residual standard error: 4.902 on 8 degrees of freedom
```

IC(μ, 95%) = **2.444** $\pm$ $t_{8,0.975}$ **1.634** = **[-1.32 , 6.21]**

Statistic: (**2.444**-0)/1.634 = **1.496** with **p-value** P(|t_8|>1.496)= pt(-1.496,8)+(1-pt(1.496,8)) = **0.173**

The assumptions of random sample and normality of the response X are assumed

- A **point estimation** of 2.44 units shows the mean discrepancy in the sample, and **95% confidence interval** for the mean is -1.32 i 6.21. P-value indicates that there is no evidence to reject 0 as a credible value
- The residual part that the model does not collect is indicated **residual standard error** (4.902), that is the standard deviation of the difference between the observed and predicted values in the sample.

Model to estimate μ. Case of μ_D in paired samples

Notation of the model : $D_i = \mu_D + \varepsilon_i$ (with D=Y1-Y2)

Functions in R: `lm(D~1)` and `summary(lm(D~1))`

(constant `1` means in R that we want to estimate only the parameter **μ**)

R outputs:

- point estimation of μ ($\hat{\mu} = \bar{d}$): **"Estimate"** of **"intercept"**
- standard error estimation: **"Std. Error"** (*se*) of **"intercept"** ($\bar{d}$). It allows to calculate CI

$$IC(1-\alpha\%, \mu) = \bar{d} \pm t_{n-1,1-\alpha/2} \cdot se$$

- value for the statistic (**"t value"**) and **p-value** (**"Pr(>|t|)"**) to evaluate the credibility for a possible parameter value (0 by default)
- standard deviation of residuals (**"Residual Standard error"**) unexplained by the model

Analysis of results

The main result is point estimation (and CI) for the population mean of the difference

In this case, it's interesting to evaluate the credibility of value 0 for the mean difference parameter. If 0 is possible then the two paired samples have the same mean parameter

The Residual standard error quantifies the deviation unexplained by them model

Model to estimate μ. Case of μ_D in paired samples

Validation of assumptions

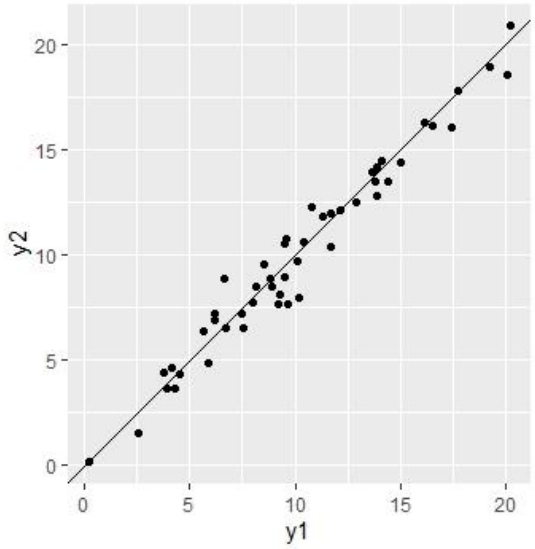
Assumptions are: **random variable** and **normality of the response**
and a constant difference between pairs **(additive efect)**

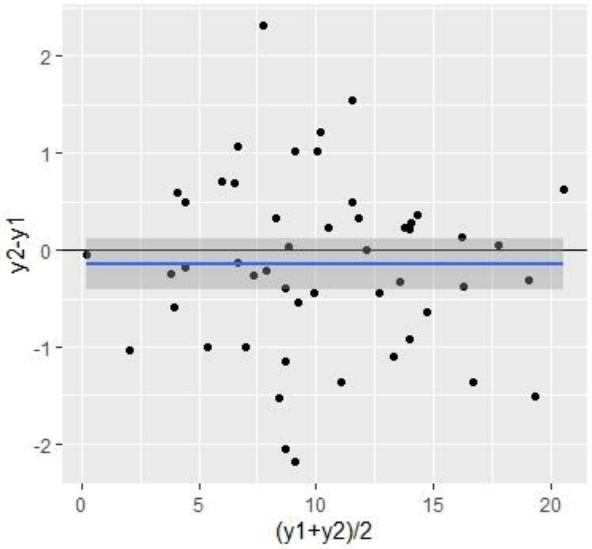
- **Random variable** assumption is not possible to verify. It depends on the sample design.
  It is only possible to assure independency in the order of the observations)
- **Normality** assumption will be verified graphically with a **Normal QQ-plot**
- An **additive effect** is assumed (e.g. an algorithm reduces 2 seconds the execution time) but sometimes the **effect is multiplicative** (e. g. an algorithm reduces by half the execution time)
  - **Bland-Altman** plot (see also in Unit C) shows the differences of the paired observations in front of the means of them. Then it is possible to study if the effect is additive or multiplicative and decide if a transformations is needed
  - **3 special situations:**
    Case 1: no lineal effect (additive)
    Case 2: multiplicative effect
    Case 3: multiplicative effect with logarithm transformation $(\ln Y1, \ln Y2)$

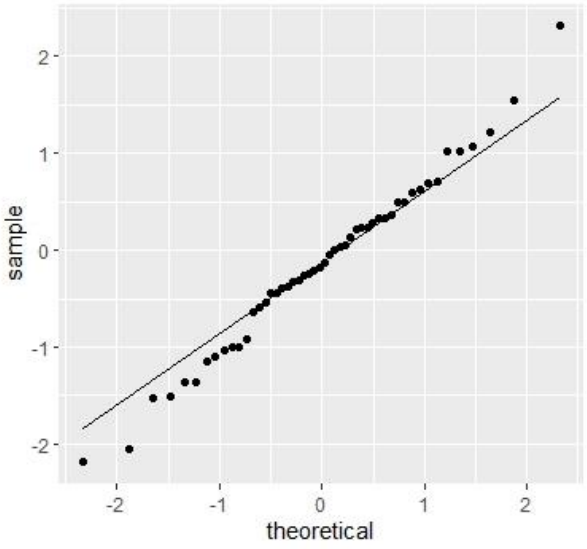
Model to estimate μ. Case of μ_D in paired samples

Validation of assumptions. Case 1: no linear effect (additive)

Example of 50 paired observations (D = Y2-Y1)







Point estimation of the mean difference is -0.146 (and p-value is "big")

CI for the mean difference is $IC_{95\%}(\mu_{Y2}-\mu_{Y1})$ = [-0.40 a 0.11] (0 is a credible value)

$Y_2 = Y_1 - 0.146$ in mean, or $Y_2 = Y_1 + [-0.40 \text{ a } 0.11]$

There is **no evidence that both means would be different**

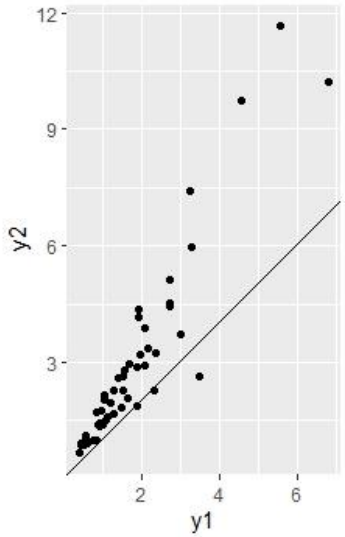
(0 value for the difference in means is credible)

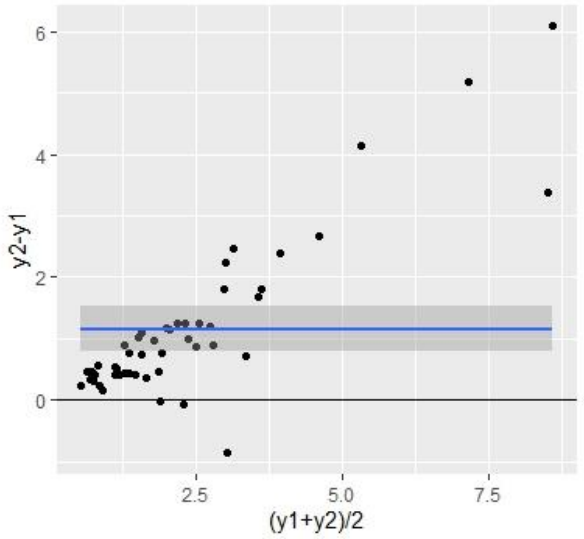
Normality of the variable difference is verified, because observed and theoretical quantiles are aligned

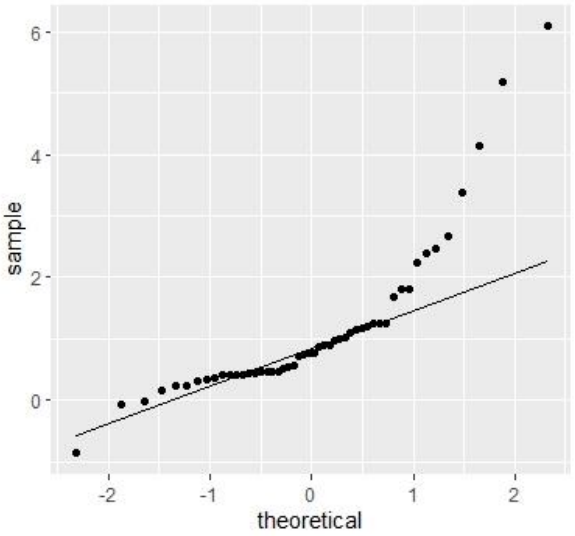
Model to estimate μ. Case of μ_D in paired samples

Validation of assumptions. Case 2: multiplicative effect

Another example of 50 paired observations







The estimation of the mean difference is 1.16 and $CI_{95\%}(\mu_{Y2}-\mu_{Y1})$ = [0.38 a 1.92]
But there **isn't a constant effect**: for small values the effect is smaller and with less variability

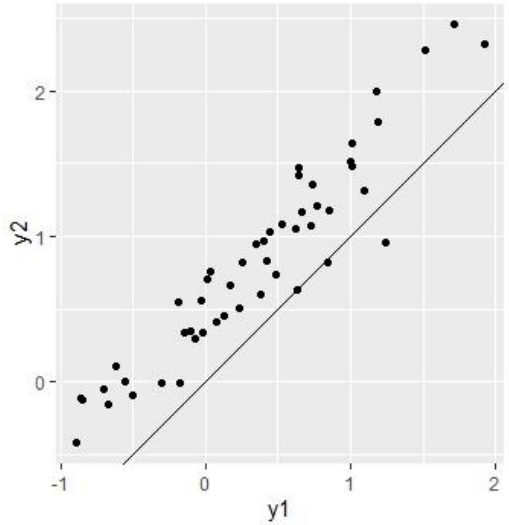
A data transformation could solve the problem → let's try with logarithm transformation

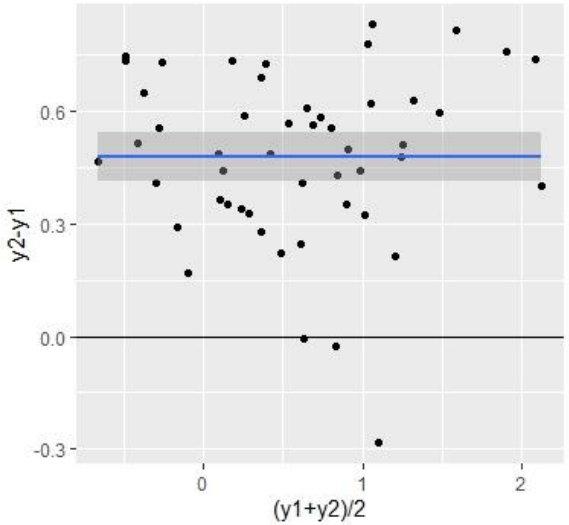
The Distribution of the differences is not normal

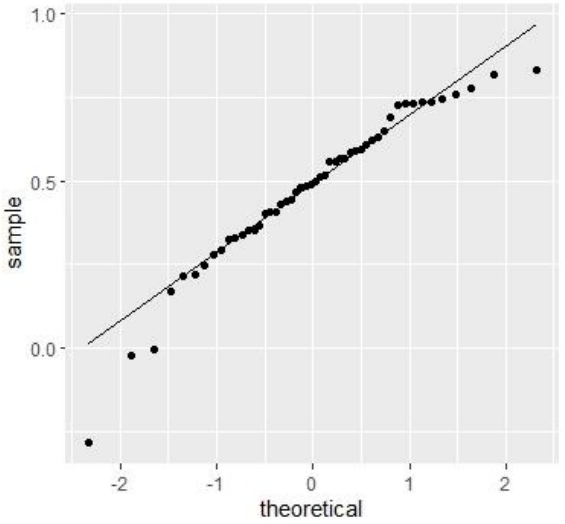
Model to estimate μ. Case of μ_D in paired samples

Validation of assumptions. Case 3: multiplicative effect and log transformation

Logarithm tranformation to previous 50 paired observations







The mean difference estimation with logarithms is 0.48, and $CI_{95\%}(\mu_{Y2'}-\mu_{Y1'})$ from 0.21 to 0.75 If Y1,Y2 are the initial variables and Y1',Y2' are the log-transformed:

$$\left.\begin{matrix} Y2' = \ln(Y2) \\ Y1' = \ln(Y1) \end{matrix}\right\} \rightarrow \quad Y2' = Y1' + 0.48 \quad (in\ mean)$$

$$\rightarrow \ln(Y2) = \ln(Y1) + 0.48 \rightarrow Y2 = e^{\ln(Y1)+0.48} \rightarrow Y2 = e^{\ln(Y1)} \cdot e^{0.48} = 1.62 \cdot Y1$$

Interpretation: Y2 is in mean 1.62 ($IC_{95\%}$ from 1.23 to 2.12) times bigger than Y1

(exp(0.21)= $e^{0.21}$ =1.23 and exp(0.75)= $e^{0.75}$ =2.12)

Normality of the variable difference of logarithms (or logarithm of the ratio) is verified, because observed and theoretical quantiles are aligned

Log transformations resulting in underestimated predicted values

Model to estimate μ. Case of μ_D in paired samples

Example of 50 paired observations (D = Y2-Y1)

Unit C solution (CI)
 	
`>t.test(Y1,Y2,paired=T)`
	`t = -1.1434 , df = 49, p-value = 0.2584`
	`alternative hypothesis: true mean is not equal to 0`
	`95 percent confidence interval:`
	`-0.4037 0.1109`
	`sample estimates: mean of x 0.146`

Unit D solution (statistical model)
 	
`> (mod <- lm(D~1))`
`> summary(mod)`
	`Coefficients:`
	`Estimate Std. Error t value Pr(>|t|)`
	`(Intercept) -0.146 0.128 -1.143 0.258`
	`Residual standard error: 0.9054 on 49 degrees of freedom`

IC(μ, 95%) = $-0.146 \pm t_{49,0.975} \cdot 0.128$ = **[-0.40 , 0.11]**

Statistic: (-**0.146**-0)/0.128 = -**1.143** with **p-valor** $P(|t_{49}|>1.143)$= pt(-1.143,49)+(1-pt(1.143,49)) = **0.258**

The assumptions of random sample and normality of the response X are assumed, and also a constant difference

- A **point estimation** of -0.146 units shows the mean difference between the paired samples., and **95% confidence interval** for the mean difference is [-0.40, 0.11]
- P-value indicates that there is o evidence to reject 0 as a credible value for the difference of means. Then the mean parameter in both paired samples could be equivalent
- The residual part that the model does not collect is the **residual standard error** (0.9054)

Statistical models

Model (b) and functions in R

(model about parameter μ in independent samples)

Model to estimate μ in independent samples

Notation for the model: $Y_i = \mu + \vartheta_k + \varepsilon_i$

(μ is the reference mean and ϑ_k is the change to the mean in group *k)*
(so, all the $\mu_k$means are not explicit parameters)

Functions in R: `lm(Y~G)` and `summary(lm(Y~G))`

(G is a column with a different character for each group or factor; if it is numerical then **as.factor(G)** is needed)

R outputs:

- point estimations of the **reference μ:** **"Estimate"** of **"intercept"**
  and estimations of the changes to means: **"Estimate"** for each change in G groups
- standard error estimation **"Std. Error"**
- value for the statistic (**"t value"**) and **p-value** (**"Pr(>|t|)"**) for each estimation that allows to evaluate the credibility for a possible parameter value (0 by default)
- standard deviation of residuals (**"Residual Standard error"**) unexplained by the model

Other functions in R (we will see in examples) to obtain complementary results:

```
confint(lm(Y~G))              # CI
library(emmeans)              # specific library to calculate means
emmeans(lm(Y~G),~G)           # means for each group
plot(emmeans(lm(Y~G),~G))     # library(ggplot2) is needed
pairs(emmeans(lm(Y~G),~G))    # to compare 2 to 2
```

Model to estimate μ in independent samples

Validation of assumptions

Assumptions are: **random variable** and **normality in each group**

and **homoscedasticity between groups** (same variability between groups)

- **Random variable** assumption is not possible to verify. It depends on the sample design. It is only possible to assure independency in the order of the observations
- **Normality** assumption will be verified graphically with a **Normal QQ-plot in each group**

- **Homoscedasticity** assumption could be verified graphically with a boxplot

Model to estimate μ in independent samples

Analysis of results

- In the model $Y_i = \mu + \vartheta_k + \varepsilon_i$ we can **separate the variability**

Total variability (T) factorize in an **explained** part (between groups or factors, E) and an unexplained part (**residual**, R**)**

In particular, these different parts of variability are defined as:

Variabilitat total (T): $SS_T = \sum_j \sum_i (y_{i,j} - \bar{y})^2$

Variabilitat residual (R) o dins dels grups: $SS_R = \sum_j \sum_i (y_{i,j} - \bar{y}_j)^2$

Variabilitat entre (E) els grups: $SS_E = \sum_j n_j (\bar{y}_j - \bar{y})^2$

- **R^2** is a ratio between the explained variability and the total variability:

$$R^2 = \frac{explained\ variability}{total\ variability} = \frac{"sum\ of\ differences\ between\ groups, squared"}{"sum\ of\ total\ differences, squared"} = \frac{SS_E}{SS_T}$$

No assumptions are required

Bigger is the R^2 value, better is the relation of the response variability and the group differences

R^2 is maximum (1=100%) if the relations is totally deterministic, residual part is 0, means in each group are very different, no differences within the groups, all the variability is between groups
R^2 is minimum (0) if the factor or group doesn't determine the variations, means in each group are equal, no variation between groups, all the variability is within groups

- R output: **"Multiple R-squared"** (and a variation, "Adjusted R-squared")

The Reading List (Statistics for Computer Engineers) offers more details in Chapter 6.6. An also in *app*

Model to estimate μ in independent samples

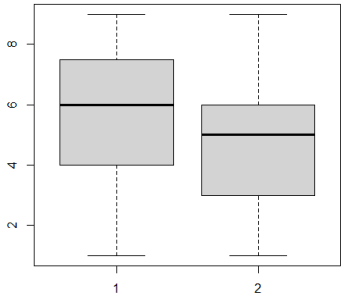
Example of 2 independent samples to compare μ_1 i μ_2

Unit C solution (CI)

```
Y1 <-c(1,2,3,5,6,6,7,7,8,8,9)          # mean=5.6  sd=2.62
Y2 <-c(1,1,3,4,5,5,6,7,9)              # mean=4.5  sd=2.65
Y <- c(1,2,3,5,6,6,7,7,8,8,9,1,1,3…)   # Y1 and Y2
G <- c(1,1,1,1,1,1,1,1,1,1,1,2,2,2…)   # in lm() as.factor(G) is needed
# mean(Y1)-mean(Y2) -> 1.080808        # 5.636364 - 4.555556 = 1.080808
```

Normality for Y1 and Y2
qqnorm(Y1) qqline(Y1)
qqnorm(Y2) i qqline(Y2)

boxplot(X1,X2)



t.test(Y1,Y2,var.equal=T) # or t.test(Y~as.factor(G),var.equal=T)

```
t = 0.91335, df = 18, p-value = 0.3731
  alternative hypothesis: true difference in means is not equal to 0
  95 percent confidence interval:
   -1.405312  3.566928
  sample estimates: mean of x mean of y  5.636364  4.555556
```

The assumptions of random sample and normality of Y1 and Y2 are assumed, and also homoscedasticity (verified graphically in the boxplot)

- The difference of the means is 1.08. P-value indicates that there is no evidence to reject 0 as a credible value for the difference of means
- The **point estimation** of 1.08 units shows the mean difference between the independent samples, and **95% confidence interval** for the mean difference is [-1.40, 3.57]
- Equality of population means could not be rejected, although there is no prove of it

Model to estimate μ in independent samples

Example of 2 independent samples to compare μ_1 i μ_2

Unit D solution (statistical model)

estimate for the reference mean is 5.6364 and for the change between groups is -1.08 (5.636 - 1.08 → 4.55)

```
lm(Y~as.factor(G))
summary(lm(Y~as.factor(G)))
```

```
Coefficients:
             Estimate   Std. Error   t value   Pr(>|t|)
(Intercept)   5.6364       0.7938     7.100    1.28e-06 ***
G2           -1.0808       1.1833    -0.913    0.373

Residual standard error: 2.633 on 18 degrees of freedom
Multiple R-squared:  0.04429,   Adjusted R-squared:  -0.008803
F-statistic: 0.8342 on 1 and 18 DF,  p-value: 0.3731
```

IC (intercept) ($\mathbf{5.6364} \pm t_{18,0.975}\ 0.7938$ = [3.97,7.30] with statistic: (**5.6364**-0) / 0.7938 = **7.1** and p-value "small"

IC ϑ_k (G2) ($\mathbf{-1.0808} \pm t_{18,0.975}\ 1.1833$ = **[-3.567,1.405]** with estatistic: (**-1.0808**-0) / 1.1833 = **-0.913** and p-value "big"

- The **point estimation** for the mean difference is -1.08
- The ratio signal/noise for the *intercept is* 7.1 (with small p-value). The reference mean (*G1*) is not credible to be equal to zero. The ratio signal/noise for the change in means is -0.9 (with big p-value), so the difference between population means could be + or -
- The residual part that the model does not collect is the **residual standard error** (**2.633**)
- **R^2** is 0.04429 (only a 4.4% of the total variability is explained by the between variability

Model to estimate μ in independent samples

Example of 2 independent samples to compare μ_1 i μ_2

Unit D solution (statistical model)

```
confint(lm(Y~as.factor(G)))
```

```
                2.5 %     97.5 %
(Intercept)  3.968623  7.304104
G2          -3.566928  1.405312
```

(CI for μ_1 and ϑ_2)

```
emmeans(lm(Y~as.factor(G),~G))
```

```
 G  emmean    SE  df  lower.CL upper.CL
 1    5.64 0.794  18      3.97      7.3
 2    4.56 0.878  18      2.71      6.4
```

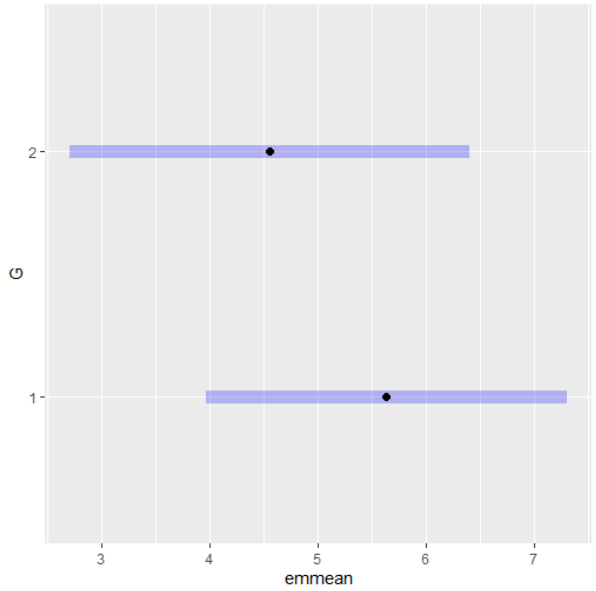
(Estimation and *se* and CI calculated)

```
pairs(emmeans(lm(Y~as.factor(G),~G))
```

```
 contrast estimate   SE df t.ratio p.value
 G1 - G2       1.08 1.18 18   0.913  0.3731
```

(Estimation and *se* and statistic calculated)

```
plot(emmeans(lm(Y~as.factor(G)),~G))
```



- The outputs of these R functions show different ways to calculate CI and statistics
- The picture show the CI for μ_1 and μ_2 with a black point as the point estimation, and a blue line for the CI range. The CI could be overlapped (meaning that is credible that μ_1 and μ_2 are equal) or not (meaning that is not credible that μ_1 and μ_2 are equal)

The results depends on the sample design: balanced design or not, that is a design with the same number of observation in each group. Balanced design is more efficient because it has lower standard error.

Statistical models
Model (c) and functions in R
(simple and multiple linear model)

Simple and multiple linear model

Notation of the model: $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ or $Y_i = \beta_0 + \beta_1 X1_i + \beta_2 X2_i + \cdots + \beta_k Xk_i + \varepsilon_i$

The parameters are the coefficients β_i of the equation

In linear model b_0 is the **intercept**, and b_1 is the slope (change in the response Y for an increase of 1 unity in the explanatory X

The coefficients fitted for the equation in case of **simple linear model** $Y = b_0 + b_1 X$ are the solution of a minimum least squares:

$b_1 = r_{XY} \cdot \frac{S_Y}{S_X} = \frac{S_{XY}}{S_X^2}$ $b_0 = \bar{Y} - b_1 \bar{X}$ and residual variance $s^2 = \frac{\sum(e_i^2)}{n-2} = \frac{(n-1)S_Y^2(1-r^2)}{n-2}$ (r_{XY} is the correlation and S_{XY} the covariance)

Variables X_i could be quantitatives or qualitatives, and we will call them predictors

Functions in R: `lm(Y~X)` and `summary(lm(Y~X))`

(or for múltiple linear model: `lm(Y~X1+X2+...)` and `summary(lm(Y~X1+X2+...))`)

R outputs:

- Point estimation (**"Estimate"**) for the "*intercept*" and for the others parameters (slope or group effect)
- standard error estimation: **"Std. Error"** that allows to calculate CI
- statistic (**"t value"**) and **p-value** (**"Pr(>|t|)"**) to evaluate the credibility for a possible parameter value (0 by default)
- standard deviation of residuals (**"Residual Standard error"**) unexplained by the model

Simple and multiple linear model

R^2: predictive capacity indicator

- In simple and multiple linear model we can **separate the variability:**

Total variability (SQ_{Total}) factorize in an **explained** part ($SQ_{Explained}$ variations in predictors X means variations in the response Y) and an unexplained part or **residual** ($SQ_{Residual}$)

In particular, these different parts of variability are defined as:

$$\sum (y_i - \bar{Y})^2 = \sum (y_i - \hat{y}_i)^2 + \sum (\hat{y}_i - \bar{Y})^2$$

$$SQ_{Total} = SQ_{Residual} + SQ_{Explicada}$$

- **R^2** is a ratio between the explained variability and the total variability:

$$R^2 = \frac{explained\ variability}{total\ variability} = \frac{SQ_E}{SQ_T}$$

No assumptions are required

Bigger is the R^2 value, better is the linear relation

R^2 is maximum (1=100%) if the relations is totally deterministic, residual part is 0
R^2 is minimum (0) if the linear model doesn't explain the response variations but the residual does

We can check in simple linear model that $R^2 = (r_{XY})^2$ that is the square of the linear correlation coefficient

The determination coefficient R^2 is a measure of the goodness of fit of the model, and r_{XY} is a measure of the association between the variables

- R output: **"Multiple R-squared"** (and a variation, "Adjusted R-squared")

The Reading List (Statistics for Computer Engineers) offers more details in Chapter 6.6

Simple and multiple linear model. Example

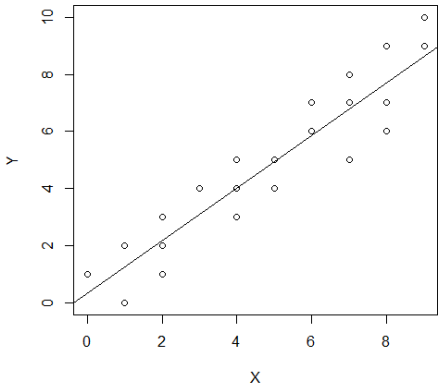
Example of a sample of a response Y and a predictor X

```
Y <-c(1,0,1,3,4,5,4,6,2,6,5,2,9,7,7,5,6,9,3,6,8,7,10,5,4,6)
X <-c(0,1,2,2,3,4,4,6,1,8,7,2,9,6,8,5,6,8,4,6,7,7,9,7,5,6)
cor(X,Y) # = 0.9239073
lm(Y~X)
summary(lm(Y~X))
```

```
 Residuals:
    Min      1Q  Median      3Q     Max
-1.7710 -0.8719  0.1483  0.8054  1.3903
Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.33581    0.44460   0.755    0.457
X            0.91931    0.07771  11.830 1.68e-11 ***

Residual standard error: 1.015 on 24 degrees of freedom
Multiple R-squared:  0.8536,    Adjusted R-squared:  0.8475
F-statistic: 139.9 on 1 and 24 DF,  p-value: 1.68e-11
```

```
plot(X,Y)
abline(lm(Y~X))
```



- The model explains **85.36 % of the total variability** of the response. The **residual** deviation is 1.015
- The **intercept** is 0.33581 with CI 0.33581±qt(0.975,24)*0.4446 -> **[-0.58,1.25]**
- The **slope** is 0.91931 with CI 0.91931±qt(0.975,24)*0.07771 -> **[0.76,1.08].** For an increase of 1 unit in X, we obtain an increase of 0.919 units in Y (between 0.76 and 1.08 with **95% confidence**)
- 0 is not a credible value for the slope, but 1 is credible for the slope. 0 is not a a credible value for intercept
  Then the equation $Y= b_0+b_1X$ could be Y=X

Simple and multiple linear model

Validation of assumptions

Assumptions are:

- **linearity** (a straight line, or a plane for the multiple model, fits the data)
- **random sample** (independence of the residuals)
- **normality of residuals**
- **Homoscedasticity of residuals** (same variability of the residuals)

Linearity concerns the deterministic part, and the other assumptions concerns the non deterministic part:

$$Y_i = \underbrace{\beta_0 + \beta_1 X1_i + \cdots \beta_k Xk_i}_{\text{Linearity}} + \underbrace{\epsilon_i}_{\text{Normality}} \rightarrow \underbrace{\epsilon_i \sim N(0,\sigma)}_{\text{Homoscedasticity}} \quad \underbrace{\epsilon_i, \epsilon_j \ ind. \forall i,j}_{\text{Independence}}$$

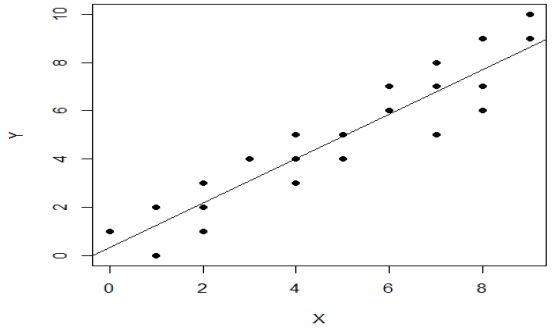
The Reading List (Statistics for Computer Engineers) offers more details in Chapter 7

Simple and multiple linear model

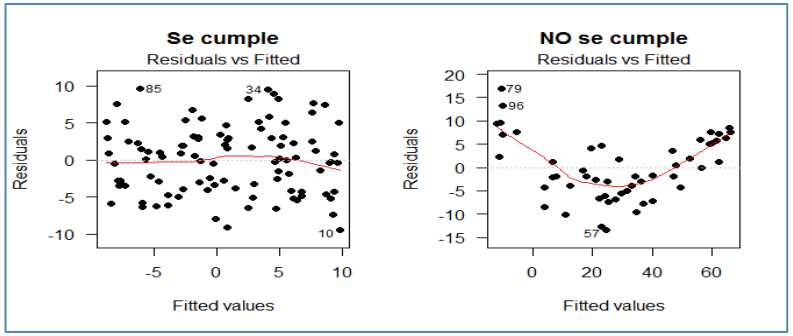
Validation of assumptions

The assumptions for linear model and their graphical verification are:

- **Linearity** (Y and X fit to a straight line)

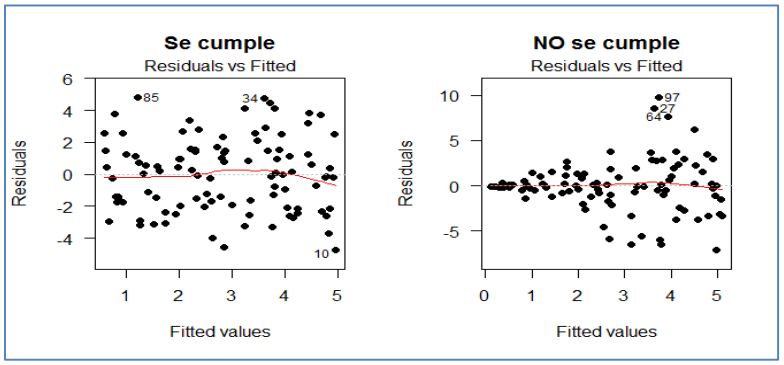


Plot Y and X to decide if data fits a straight line



Residual pictures to decide if data fits an horizontal line

- **Homoscedasticity** ($\varepsilon \sim N(0, \sigma)$ or a constant variability)

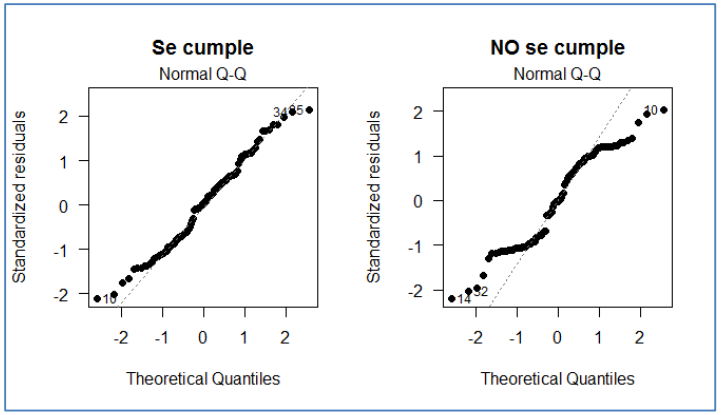


Picture of residuals in front of predictions to decide if they have the same variability all over the range of values

Simple and multiple linear model

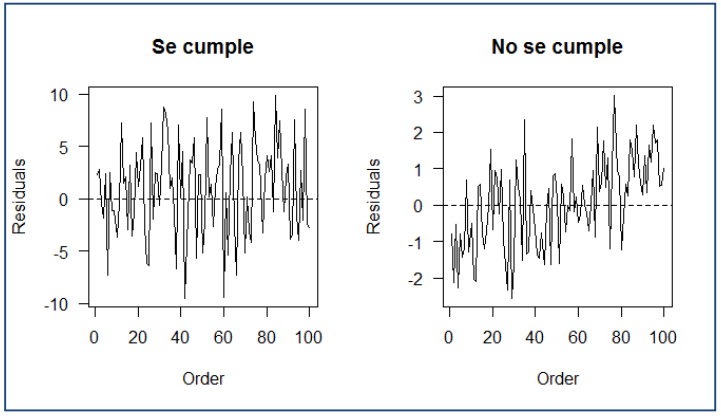
Validation of assumptions

- **Normality** ($\varepsilon \sim N(0, \sigma)$)



Qqnorm() and qqline() to decide if the residuals fit the Normal model

- **Independence** (random sample and non dependency between observations)



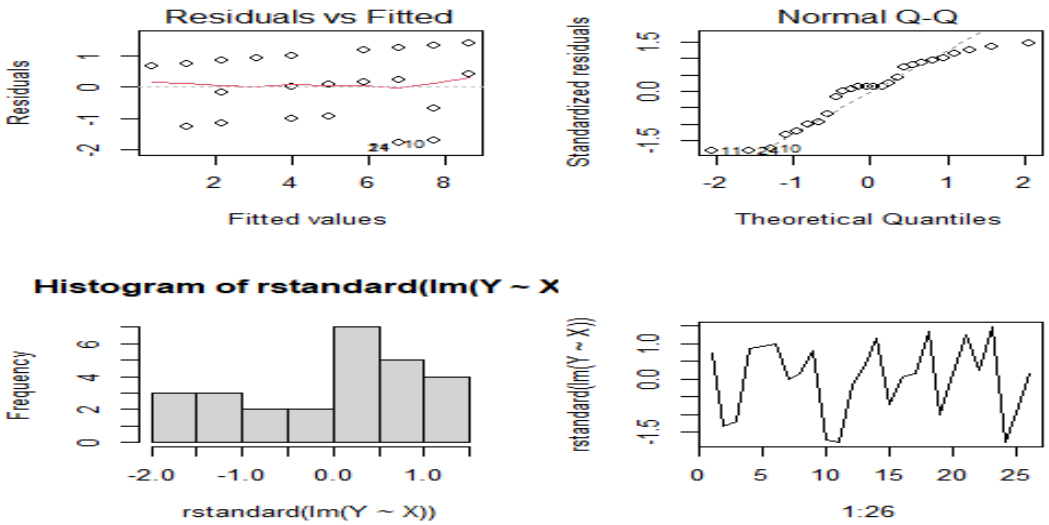
Residual independence depends on the sample design. It is only possible to assure independency in the order of the observations. That is we can obtain the picture of residuals by order and decide if there is a pattern (dependency) or not (independency)

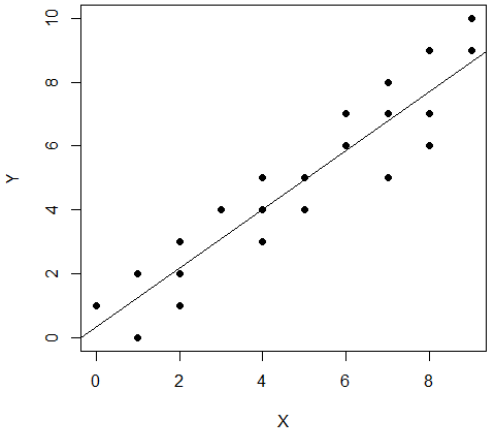
Simple and multiple linear model

Example of a sample of a response Y and a predictor X

```
Y <-c(1,0,1,3,4,5,4,6,2,6,5,2,9,7,7,5,6,9,3,6,8,7,10,5,4,6)
X <-c(0,1,2,2,3,4,4,6,1,8,7,2,9,6,8,5,6,8,4,6,7,7,9,7,5,6)
cor(X,Y) # = 0.9239073
lm(Y~X)
par(mfrow=c(2,2))
plot(lm(Y~X),c(2,1)) # o bé plot(model,c(2,3))
hist(rstandard(lm(Y~X)))
plot(rstandard(lm(Y~X)),type="l")
```

```
plot(X,Y)
abline(lm(Y~X))
```





All the assumptions are verified, except Normality that could be questionable in the extremes

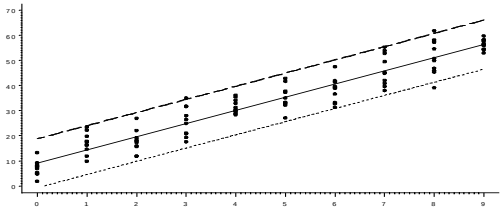
Simple and multiple linear model

Analysis of results (Functions in R to predict)

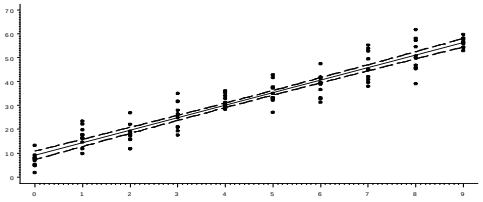
- For a point prediction we can calculate CI of two types:
  - **Individual value**. We want to estimate the **value** for a **specific** observation
    `predict(…, interval = "prediction")`
  - **Expected value**. We want to estimate the **mean** for a **all** observations of a type
    `predict(…, interval = "confidence")`

Both cases have the same **point predicted estimation but not the uncertainty** (for expected value the CI is narrow)

(individual value prediction)



(expected value prediction)



- Function in R:

```
predict(lm( ))  # point prediction
new <- data.frame([column_name]=[value for pred],...)
predict(lm( ),new ) # point prediction to new obs
predict(lm( ),new,interval=...) # 95%CI
predict(lm( ),new,interval=...,level=...)
```

Predictions for the same observations observacions.
New observations to predict
Predictions for new observations
Interval prediction
Interval prediction for a specific level of confidence

The Reading List (Statistics for Computer Engineers) offers more details in Chapter 7.2

Multiple linear model. Example

Example of a model using quantitatives and qualitatives explanatory variables

Statistical models usually combine (b) and (c) models, but more variables imply additional difficulties, mainly if they aren't independent

The gender pay gap is the salary differences between man and women. Here there is the output result, for a sample of 30 men and 20 women, of the model with salary as the response and experience, age or gender as explanatories variables.

Y: anual salary (thousand of eur) (is the response variable)

G: gender ("m", man, "w", woman)

Xp: number of working years

Age: years old

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	**44.0387**	**10.9254**	**4.031**	0.000207
xp	**1.1575**	**0.5638**	**2.053**	0.045789
age	**-0.3088**	**0.4683**	**-0.659**	0.512878
Gw	**-5.2430**	**0.8572**	**-6.116**	1.94e-07

Residual standard error: 2.663 on 46 degr of freedom

Multiple R-squared: 0.7647, Adjusted R-squared: 0.7494

- "Estimate" is the **point estimation** of the slope for quantitative variables, or the group effect for the qualitative variables
- "*Intercept*" is the estimation of the mean response for reference categories ("man") and for the value 0 of quantitative variables
- "Estimates" show the variable change (increase of 1 in case of quantitative variable) respect the reference values, *assuming no change for others covariables* (not usual because of correlations)
- In general, we shouldn't apply experimental designs to obtain independent data

Multiple linear model. Example

Example of a model using quantitatives and qualitatives explanatory variables

The gender pay gap is the salary differences between man and women. Here there is the output result, for a sample of 30 men and 20 women, of the model with salary as the response and experience, age or gender as explanatories variables.

Y: anual salary (thousand of eur) (is the response variable)

G: gender ("m", man, "w", woman)

Xp: number of working years

Age: years old

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)  44.0387    10.9254   4.031 0.000207
xp            1.1575     0.5638   2.053 0.045789
age          -0.3088     0.4683  -0.659 0.512878
Gw           -5.2430     0.8572  -6.116 1.94e-07
Residual standard error: 2.663 on 46 degr of freedom
Multiple R-squared:  0.7647,  Adjusted R-squared:  0.7494
```

- "*Intercept*" is the estimation of the mean response for reference categories ("man") and for the value 0 of quantitative variables: **44039 €** as a base salary, and **10925€** is the standard error
- Coefficients fitted for each variable (for each case when the others are fixed) mean:
  - one more working year, "represents"* an increase of **1157.5 €** in salary
  - one more year old, a decrease of **308.8 €** in salary A large se and correlation, or collinearity, between age and experience is difficult to interpret:
  - In case of a woman, a decreases of **5243 €** in salary
  - There is no data, but if gender is not independent of experience, both estimations are mislead
  - The model is useful for global predictions, but not particular contributions

(* "represents" in the sense of "association", not "causality")

Other techniques

Other techniques



Two types of models (L. Breiman, Two cultures. 2001)

Models for Explanation (theory driven).

- Models for understanding the **true generative mechanism** of data (this implies that we look for the **causal relationships** of the response).
- Imply **interpretable** and **parsimonious** models. This leads using parametric (statistical) models. Error follows a probability distribution.
- We are interested in **significance**. We focus on global measures of fit and significance of coefficients (we use p.values).
- Prediction may allow **forecasting the future** in presence of change (or intervention policies).
- Modeling needs expert domain.

Models for Prediction (data driven).

- Models are considered mere **algorithms**.
- The model is considered a **black box** (complexity is not a problem).
- **Correlated** predictors are enough.
- We are just interested in the **accuracy** of predictions (significance is uninteresting).
- We focus on the **Generalization Error** of the model. Error should be made minimal (we don't care on the residual analysis).
- Predictions **fail in presence of change**.
- Accuracy and interpretability are in conflict.

Descriptive Analytics
discovering patterns in data

- Profiling
- Principal Component Analysis
- Correspodence Analysis
- Clustering
- Association Rules
- ...

Predictive Analytics
Setting models for prediction

- Multiple Regression
- Logistic Regression (GLM)
- Neural networks
- Decision trees
- Knn neighbors
- PCR & PLSR
- Discriminant Analysis
- SVM
- Deep Learning

Other techniques

Techniques in data mining, data science, artificial intelligence are different from classical statistical techniques in several aspects:

- Usually they are observational studies rather than experimental
- Usually they use massive data without a previous design to assure randomness
- They use specific measures of goodness or predictive capacity (but usually they are % of explained variability like R^2 in classical statistics)
- They don't use parametric techniques and they provide mainly graphical outputs
- Many techniques are iterative algorithms not searching a model, computationally demanding but more and more suitable in case of huge data

Functions in R:

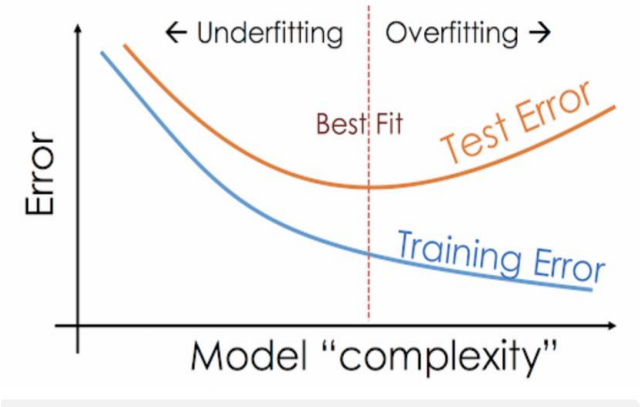
- Multidimensional visualization techniques `PCA()`
- Clustering techniques `HCPC()` `kmeans()`

(some explanations and examples could be found at the end of the exercises for unit D)

There are a lot of different techniques (`randomForest()`, `neuralnet()`, …) but is very important to assess the predictive capacity and detect possible problems of overfitting (as we will explain in the next slide)

Predictive capacity and overfitting

- Large data (to analyse with data science techniques but also with simple and mainly multiple linear model) can afford to assess **predictive capacity** splitting the sample in a **training** data set and a **test** data set (usually 70% and 30%)
  - **training** data set is used to fit the model
  - **test** data set is used to evaluate the model
    (we want to evaluate the model in an independent data set, different from the one that fitted the model)
  - Predictive capacity should measure an "**error**" (minimise some kind of difference between observed and predicted values)
- **Overfitting** occurs when a model fits good for the training data set but not for the test data set
  - An overfitted model will generate poor predictions for future observations
  - Sometimes this is because a complex model could be very accurate for the training data but not so adjusted for differences in new future data. The multiple linear model usually use too many predictor variables without a real relation to the response variable that could involve overfitting



Usually as the "complexity" of a model increases the error decreases for training and test data sets. But it could happen that more complexity achive small error in training data sets but not In test data sets (and also in future data).
The best model is just before overfitting happens